Table 1: Simulation results under retrospective sampling with good overlap ($\psi = 0.6$): signed relative bias (%), relative efficiency, and coverage of 95% Wald confidence intervals across 1,000 replications.

Estimand	Method	Score Model		Signed Relative Bias (%)				Relative Efficiency				Coverage (%)			
		PS	PGS	Cor—Cor	Mis—Cor	Cor—Mis	Mis—Mis	Cor—Cor	Mis—Cor	Cor—Mis	Mis—Mis	Cor—Cor	Mis—Cor	Cor—Mis	Mis—Mis
PATE	PSM	×	–	6.4	14.7	6.4	14.7	1.00	0.85	1.00	0.85	97.0	84.0	97.0	84.0
	PGM	–	×	6.3	6.3	11.4	11.4	2.60	2.60	1.82	1.82	81.2	81.2	68.5	68.5
	DSM	×	×	6.4	6.4	6.6	10.5	2.35	2.34	2.11	1.85	81.6	81.5	87.6	70.9
	SW-PSM	✓	–	−0.1	−0.4	−0.1	−0.4	1.75	1.70	1.75	1.70	93.2	92.6	93.2	92.6
	WDSM (M=1)	✓	×	−0.2	0.0	−2.7	4.6	1.85	1.82	1.51	1.17	94.2	94.5	96.0	91.5
	WDSM (M=3)	✓	×	−0.2	−0.1	−0.8	5.9	2.26	2.23	1.84	1.36	94.3	94.7	96.2	89.1
	WDSM (M=5)	✓	×	−0.2	−0.1	−0.4	6.3	2.33	2.30	1.93	1.42	94.6	94.3	96.7	87.9
PATT	PSM	×	–	6.0	11.4	6.0	11.4	1.00	0.86	1.00	0.86	96.1	83.5	96.1	83.5
	PGM	–	×	5.8	5.8	12.9	12.9	1.98	1.98	1.32	1.32	80.3	80.3	53.1	53.1
	DSM	×	×	5.7	5.7	5.6	9.8	1.89	1.93	1.46	1.31	81.2	80.4	87.2	69.9
	SW-PSM	✓	–	−0.2	−0.4	−0.2	−0.4	2.19	2.25	2.19	2.25	90.9	89.0	90.9	89.0
	WDSM (M=1)	✓	×	−0.2	−0.2	0.2	3.6	1.88	1.84	1.57	1.40	93.7	94.0	96.2	91.0
	WDSM (M=3)	✓	×	−0.3	−0.2	0.0	3.9	2.29	2.25	1.98	1.59	94.1	94.2	96.8	88.6
	WDSM (M=5)	✓	×	−0.3	−0.3	−0.1	4.0	2.38	2.35	2.09	1.67	94.9	94.3	97.0	88.7

Note: PS = propensity score model; PGS = prognostic score model. ✓ denotes survey-weighted score estimation, × unweighted estimation, and – a score not used by that matching method. WDSM uses survey-weighted propensity estimation under retrospective sampling; prognostic fits are unweighted. WDSM additionally uses survey weights in matched imputation, bias correction, and Hájek normalization. Correct models include all six main effects and the X_1X_2 interaction; misspecified models omit that interaction. The first/second specification labels refer to PS/PGS. SW-PSM retains an always-correct treatment-interacted outcome regression. Signed relative bias is $100(\widehat{\tau} - \tau)/|\tau|$. Relative efficiency is the Monte Carlo variance of PSM under Cor—Cor divided by that of each method within the same estimand, sampling design, and overlap. PSM, PGM, and DSM use $M = 1$ matching with replacement; SW-PSM uses optimal full matching; WDSM uses $M \in \{1, 3, 5\}$. All intervals are centered normal Wald intervals. WDSM uses fixed-original-reuse replication variance with divisor B; PSM, PGM, and DSM retain the author's bootstrap sample variance with divisor $B - 1$; SW-PSM uses subclass-cluster sandwich/delta-method variance. Bootstrap methods use $B = 200$. All requested records passed the strict summary audit; no failed records are silently omitted.

Table 2: Simulation results under retrospective sampling with poor overlap ($\psi = 2.0$): signed relative bias (%), relative efficiency, and coverage of 95% Wald confidence intervals across 1,000 replications.

Estimand	Method	Score Model		Signed Relative Bias (%)				Relative Efficiency				Coverage (%)			
		PS	PGS	Cor—Cor	Mis—Cor	Cor—Mis	Mis—Mis	Cor—Cor	Mis—Cor	Cor—Mis	Mis—Mis	Cor—Cor	Mis—Cor	Cor—Mis	Mis—Mis
PATE	PSM	×	–	10.5	38.1	10.5	38.1	1.00	1.06	1.00	1.06	94.8	78.4	94.8	78.4
	PGM	–	×	6.1	6.1	22.7	22.7	22.01	22.01	13.67	13.67	87.9	87.9	39.1	39.1
	DSM	×	×	5.8	6.0	6.1	16.9	7.90	7.93	6.98	6.58	93.3	93.0	94.2	79.5
	SW-PSM	✓	–	−1.0	−1.2	−1.0	−1.2	3.22	3.18	3.22	3.18	81.3	83.8	81.3	83.8
	WDSM (M=1)	✓	×	0.3	0.6	−0.6	24.2	6.67	6.73	4.43	3.48	95.4	94.6	95.4	77.5
	WDSM (M=3)	✓	×	0.7	0.7	2.6	26.3	8.69	8.70	5.95	4.42	94.4	94.9	95.8	67.9
	WDSM (M=5)	✓	×	0.9	0.9	3.8	26.7	9.25	9.32	6.57	4.88	94.1	94.4	95.5	63.6
PATT	PSM	×	–	4.5	21.0	4.5	21.0	1.00	0.91	1.00	0.91	96.1	80.8	96.1	80.8
	PGM	–	×	3.4	3.4	22.6	22.6	14.11	14.11	8.96	8.96	90.1	90.1	7.1	7.1
	DSM	×	×	3.6	3.5	4.7	17.5	5.02	5.30	3.33	3.46	91.3	92.8	91.6	57.5
	SW-PSM	✓	–	−0.6	−0.8	−0.6	−0.8	4.47	4.72	4.47	4.72	85.4	86.9	85.4	86.9
	WDSM (M=1)	✓	×	0.2	0.1	2.0	13.0	7.66	7.62	5.13	5.00	94.8	94.5	95.1	68.3
	WDSM (M=3)	✓	×	0.3	0.2	2.6	13.7	9.41	9.25	6.67	6.48	94.8	93.8	95.1	58.2
	WDSM (M=5)	✓	×	0.3	0.1	2.8	13.9	9.86	9.69	6.98	6.86	94.9	94.2	94.8	54.7

Note: PS = propensity score model; PGS = prognostic score model. ✓ denotes survey-weighted score estimation, × unweighted estimation, and – a score not used by that matching method. WDSM uses survey-weighted propensity estimation under retrospective sampling; prognostic fits are unweighted. WDSM additionally uses survey weights in matched imputation, bias correction, and Hájek normalization. Correct models include all six main effects and the X_1X_2 interaction; misspecified models omit that interaction. The first/second specification labels refer to PS/PGS. SW-PSM retains an always-correct treatment-interacted outcome regression. Signed relative bias is $100(\widehat{\tau} - \tau)/|\tau|$. Relative efficiency is the Monte Carlo variance of PSM under Cor—Cor divided by that of each method within the same estimand, sampling design, and overlap. PSM, PGM, and DSM use $M = 1$ matching with replacement; SW-PSM uses optimal full matching; WDSM uses $M \in \{1, 3, 5\}$. All intervals are centered normal Wald intervals. WDSM uses fixed-original-reuse replication variance with divisor B; PSM, PGM, and DSM retain the author's bootstrap sample variance with divisor $B - 1$; SW-PSM uses subclass-cluster sandwich/delta-method variance. Bootstrap methods use $B = 200$. All requested records passed the strict summary audit; no failed records are silently omitted.

Table 3: Simulation results under prospective sampling with good overlap ($\psi = 0.6$): signed relative bias (%), relative efficiency, and coverage of 95% Wald confidence intervals across 1,000 replications.

Estimand	Method	Score Model		Signed Relative Bias (%)				Relative Efficiency				Coverage (%)			
		PS	PGS	Cor—Cor	Mis—Cor	Cor—Mis	Mis—Mis	Cor—Cor	Mis—Cor	Cor—Mis	Mis—Mis	Cor—Cor	Mis—Cor	Cor—Mis	Mis—Mis
PATE	PSM	×	–	6.4	14.6	6.4	14.6	1.00	0.87	1.00	0.87	95.8	82.4	95.8	82.4
	PGM	–	×	5.9	5.9	11.4	11.4	2.35	2.35	1.78	1.78	81.3	81.3	67.6	67.6
	DSM	×	×	5.9	5.8	6.3	10.5	2.30	2.32	2.00	1.76	81.8	83.2	86.0	72.1
	SW-PSM	✓	–	−0.6	−0.7	−0.6	−0.7	1.89	1.88	1.89	1.88	92.5	91.7	92.5	91.7
	WDSM (M=1)	×	×	−0.7	−0.8	−1.2	5.8	2.12	2.13	1.65	1.34	92.6	93.1	95.9	88.5
	WDSM (M=3)	×	×	−0.6	−0.6	−0.4	6.4	2.30	2.31	1.92	1.48	92.2	92.9	96.3	87.9
	WDSM (M=5)	×	×	−0.6	−0.6	−0.3	6.4	2.37	2.36	1.92	1.49	93.2	93.3	96.4	87.2
PATT	PSM	×	–	5.9	11.4	5.9	11.4	1.00	0.81	1.00	0.81	95.7	82.8	95.7	82.8
	PGM	–	×	5.4	5.4	12.8	12.8	1.73	1.73	1.15	1.15	80.7	80.7	52.7	52.7
	DSM	×	×	5.6	5.5	5.4	10.0	1.74	1.72	1.43	1.14	81.4	80.8	89.1	67.1
	SW-PSM	✓	–	−0.7	−0.6	−0.7	−0.6	1.57	1.57	1.57	1.57	90.5	90.3	90.5	90.3
	WDSM (M=1)	×	×	−0.5	−0.5	−2.1	2.5	1.36	1.34	1.15	0.97	93.4	93.4	96.7	93.1
	WDSM (M=3)	×	×	−0.6	−0.6	−1.3	3.0	1.58	1.58	1.35	1.15	93.7	94.0	95.8	92.0
	WDSM (M=5)	×	×	−0.6	−0.7	−1.0	3.2	1.62	1.64	1.42	1.18	93.8	94.5	95.9	90.6

Note: PS = propensity score model; PGS = prognostic score model. ✓ denotes survey-weighted score estimation, × unweighted estimation, and – a score not used by that matching method. WDSM uses unweighted propensity estimation under prospective sampling; prognostic fits are unweighted. WDSM additionally uses survey weights in matched imputation, bias correction, and Hájek normalization. Correct models include all six main effects and the X_1X_2 interaction; misspecified models omit that interaction. The first/second specification labels refer to PS/PGS. SW-PSM retains an always-correct treatment-interacted outcome regression. Signed relative bias is $100(\widehat{\tau} - \tau)/|\tau|$. Relative efficiency is the Monte Carlo variance of PSM under Cor—Cor divided by that of each method within the same estimand, sampling design, and overlap. PSM, PGM, and DSM use $M = 1$ matching with replacement; SW-PSM uses optimal full matching; WDSM uses $M \in \{1, 3, 5\}$. All intervals are centered normal Wald intervals. WDSM uses fixed-original-reuse replication variance with divisor B; PSM, PGM, and DSM retain the author's bootstrap sample variance with divisor $B - 1$; SW-PSM uses subclass-cluster sandwich/delta-method variance. Bootstrap methods use $B = 200$. All requested records passed the strict summary audit; no failed records are silently omitted.

Table 4: Simulation results under prospective sampling with poor overlap ($\psi = 2.0$): signed relative bias (%), relative efficiency, and coverage of 95% Wald confidence intervals across 1,000 replications.

Estimand	Method	Score Model		Signed Relative Bias (%)				Relative Efficiency				Coverage (%)			
		PS	PGS	Cor—Cor	Mis—Cor	Cor—Mis	Mis—Mis	Cor—Cor	Mis—Cor	Cor—Mis	Mis—Mis	Cor—Cor	Mis—Cor	Cor—Mis	Mis—Mis
PATE	PSM	×	–	10.6	39.7	10.6	39.7	1.00	0.91	1.00	0.91	96.3	82.5	96.3	82.5
	PGM	–	×	5.6	5.6	23.4	23.4	21.74	21.74	12.82	12.82	89.1	89.1	48.9	48.9
	DSM	×	×	5.3	5.4	5.6	17.3	7.70	7.71	6.84	6.34	92.0	92.1	93.4	82.4
	SW-PSM	✓	–	−1.5	−1.1	−1.5	−1.1	4.57	4.42	4.57	4.42	82.8	83.3	82.8	83.3
	WDSM (M=1)	×	×	−1.0	−1.2	−0.6	23.9	9.28	9.11	6.49	5.45	94.5	94.5	96.5	74.9
	WDSM (M=3)	×	×	−0.3	−0.4	1.8	25.5	10.97	10.83	7.90	6.22	94.0	94.5	96.2	66.9
	WDSM (M=5)	×	×	0.0	−0.1	2.7	25.8	11.66	11.61	8.34	6.43	94.5	94.7	96.2	65.1
PATT	PSM	×	–	6.7	24.3	6.7	24.3	1.00	0.90	1.00	0.90	95.7	71.5	95.7	71.5
	PGM	–	×	5.0	5.0	24.0	24.0	11.62	11.62	8.02	8.02	81.9	81.9	5.2	5.2
	DSM	×	×	4.8	4.7	5.5	18.7	4.50	4.64	2.94	3.01	89.2	90.2	89.2	54.2
	SW-PSM	✓	–	−1.2	−1.0	−1.2	−1.0	2.96	2.91	2.96	2.91	84.0	81.6	84.0	81.6
	WDSM (M=1)	×	×	−0.7	−0.6	−0.8	11.2	4.89	4.90	3.12	3.14	93.8	93.4	95.5	78.2
	WDSM (M=3)	×	×	−0.4	−0.5	0.9	12.3	5.57	5.55	3.75	3.69	93.9	94.3	94.1	71.2
	WDSM (M=5)	×	×	−0.3	−0.4	1.6	12.6	5.75	5.65	3.88	3.79	93.1	93.5	93.4	70.4

Note: PS = propensity score model; PGS = prognostic score model. ✓ denotes survey-weighted score estimation, × unweighted estimation, and – a score not used by that matching method. WDSM uses unweighted propensity estimation under prospective sampling; prognostic fits are unweighted. WDSM additionally uses survey weights in matched imputation, bias correction, and Hájek normalization. Correct models include all six main effects and the X_1X_2 interaction; misspecified models omit that interaction. The first/second specification labels refer to PS/PGS. SW-PSM retains an always-correct treatment-interacted outcome regression. Signed relative bias is $100(\widehat{\overline{\tau}} - \tau)/|\tau|$. Relative efficiency is the Monte Carlo variance of PSM under Cor—Cor divided by that of each method within the same estimand, sampling design, and overlap. PSM, PGM, and DSM use $M = 1$ matching with replacement; SW-PSM uses optimal full matching; WDSM uses $M \in \{1, 3, 5\}$. All intervals are centered normal Wald intervals. WDSM uses fixed-original-reuse replication variance with divisor B; PSM, PGM, and DSM retain the author's bootstrap sample variance with divisor $B - 1$; SW-PSM uses subclass-cluster sandwich/delta-method variance. Bootstrap methods use $B = 200$. All requested records passed the strict summary audit; no failed records are silently omitted.

Table 5: Retrospective good overlap: paired point-estimator relative-bias comparison across 1000 sampled data sets.

Estimand	M	PS/PG	Original RB (%)	Stabilized RB (%)	ARB reduction (pp)	MC error interval (pp)	Failed/N	MC SD original/stabilized	RMSE original/stabilized
PATE	1	Cor—Cor	-0.18	-0.18	-0.01	[-0.70, 0.69]	0/1000	0.062 / 0.063	0.062 / 0.063
PATE	1	Mis—Cor	0.03	0.02	0.01	[-0.54, 0.54]	0/1000	0.063 / 0.063	0.063 / 0.063
PATE	1	Cor—Mis	-2.70	-0.04	2.66	[1.52, 3.26]	0/1000	0.069 / 0.071	0.073 / 0.071
PATE	1	Mis—Mis	4.64	6.69	-2.04	[-3.34, -0.74]	0/1000	0.078 / 0.081	0.088 / 0.099
PATE	3	Cor—Cor	-0.17	-0.19	-0.02	[-0.65, 0.63]	0/1000	0.056 / 0.057	0.056 / 0.057
PATE	3	Mis—Cor	-0.13	-0.15	-0.01	[-0.62, 0.60]	0/1000	0.057 / 0.057	0.057 / 0.057
PATE	3	Cor—Mis	-0.81	0.16	0.65	[-0.38, 1.32]	0/1000	0.062 / 0.063	0.063 / 0.063
PATE	3	Mis—Mis	5.93	6.69	-0.76	[-1.96, 0.44]	0/1000	0.073 / 0.074	0.089 / 0.094
PATE	5	Cor—Cor	-0.17	-0.18	-0.01	[-0.64, 0.62]	0/1000	0.055 / 0.056	0.055 / 0.056
PATE	5	Mis—Cor	-0.11	-0.12	-0.01	[-0.59, 0.57]	0/1000	0.056 / 0.056	0.056 / 0.056
PATE	5	Cor—Mis	-0.37	0.22	0.15	[-0.72, 0.87]	0/1000	0.061 / 0.062	0.061 / 0.062
PATE	5	Mis—Mis	6.27	6.73	-0.46	[-1.63, 0.71]	0/1000	0.071 / 0.072	0.089 / 0.092
PATT	1	Cor—Cor	-0.25	-0.27	-0.02	[-0.68, 0.65]	0/1000	0.059 / 0.060	0.059 / 0.060
PATT	1	Mis—Cor	-0.22	-0.26	-0.03	[-0.67, 0.64]	0/1000	0.060 / 0.060	0.060 / 0.060
PATT	1	Cor—Mis	0.18	-0.37	-0.19	[-0.81, 0.62]	0/1000	0.065 / 0.064	0.065 / 0.064
PATT	1	Mis—Mis	3.57	4.07	-0.50	[-1.44, 0.44]	0/1000	0.068 / 0.069	0.078 / 0.081
PATT	3	Cor—Cor	-0.25	-0.27	-0.01	[-0.64, 0.62]	0/1000	0.054 / 0.054	0.054 / 0.054
PATT	3	Mis—Cor	-0.25	-0.26	-0.01	[-0.63, 0.62]	0/1000	0.054 / 0.054	0.054 / 0.054
PATT	3	Cor—Mis	0.05	-0.25	-0.20	[-0.65, 0.44]	0/1000	0.058 / 0.058	0.058 / 0.058
PATT	3	Mis—Mis	3.94	3.96	-0.02	[-0.91, 0.86]	0/1000	0.064 / 0.065	0.076 / 0.077
PATT	5	Cor—Cor	-0.34	-0.35	-0.01	[-0.71, 0.70]	0/1000	0.053 / 0.053	0.053 / 0.053
PATT	5	Mis—Cor	-0.29	-0.29	-0.01	[-0.66, 0.65]	0/1000	0.053 / 0.053	0.053 / 0.053
PATT	5	Cor—Mis	-0.08	-0.26	-0.19	[-0.65, 0.46]	0/1000	0.056 / 0.056	0.056 / 0.056
PATT	5	Mis—Mis	3.98	3.98	0.01	[-0.85, 0.87]	0/1000	0.063 / 0.063	0.075 / 0.075

Relative bias (RB) is 100*(mean estimate - truth)/abs(truth). Absolute relative bias (ARB) is abs(RB), not mean absolute error. ARB reduction is original ARB minus stabilized ARB; positive favors stabilization. MC interval is the protocol's conservative approximate 95% Monte Carlo error interval for bias reduction within each setting. It controls the two mean-bias intervals within that setting; there is no adjustment across the 96 settings. This is not a confidence interval for a stabilized treatment effect. SD denotes Monte Carlo SD. NA means full-run stabilized performance is unavailable because at least one requested estimate failed. No conditional-on-success metric enters these primary outputs. Requested paired replications: 1000.

Table 6: Retrospective poor overlap: paired point-estimator relative-bias comparison across 1000 sampled data sets.

Estimand	M	PS/PG	Original RB (%)	Stabilized RB (%)	ARB reduction (pp)	MC error interval (pp)	Failed/N	MC SD original/stabilized	RMSE original/stabilized
PATE	1	Cor—Cor	0.35	0.38	-0.03	[-1.44, 1.34]	0/1000	0.122 / 0.130	0.122 / 0.130
PATE	1	Mis—Cor	0.58	0.64	-0.05	[-1.69, 1.58]	0/1000	0.121 / 0.129	0.121 / 0.129
PATE	1	Cor—Mis	-0.55	3.23	-2.68	[-4.57, -0.12]	0/1000	0.149 / 0.163	0.149 / 0.165
PATE	1	Mis—Mis	24.24	28.03	-3.79	[-6.69, -0.88]	0/1000	0.169 / 0.186	0.269 / 0.306
PATE	3	Cor—Cor	0.71	0.76	-0.05	[-1.66, 1.59]	0/1000	0.107 / 0.110	0.107 / 0.110
PATE	3	Mis—Cor	0.75	0.81	-0.06	[-1.71, 1.62]	0/1000	0.107 / 0.110	0.107 / 0.110
PATE	3	Cor—Mis	2.65	3.96	-1.31	[-3.47, 0.85]	0/1000	0.129 / 0.134	0.131 / 0.139
PATE	3	Mis—Mis	26.27	27.64	-1.37	[-3.88, 1.13]	0/1000	0.150 / 0.156	0.272 / 0.286
PATE	5	Cor—Cor	0.87	0.92	-0.04	[-1.76, 1.67]	0/1000	0.103 / 0.106	0.104 / 0.106
PATE	5	Mis—Cor	0.92	0.99	-0.07	[-1.78, 1.63]	0/1000	0.103 / 0.105	0.103 / 0.105
PATE	5	Cor—Mis	3.76	4.56	-0.80	[-2.83, 1.24]	0/1000	0.123 / 0.126	0.127 / 0.132
PATE	5	Mis—Mis	26.66	27.45	-0.79	[-3.15, 1.57]	0/1000	0.142 / 0.146	0.271 / 0.279
PATT	1	Cor—Cor	0.24	0.18	0.06	[-0.67, 0.75]	0/1000	0.085 / 0.082	0.085 / 0.082
PATT	1	Mis—Cor	0.09	0.01	0.08	[-0.50, 0.60]	0/1000	0.085 / 0.082	0.085 / 0.082
PATT	1	Cor—Mis	1.97	2.50	-0.53	[-1.74, 0.68]	0/1000	0.104 / 0.098	0.107 / 0.102
PATT	1	Mis—Mis	13.04	14.01	-0.96	[-2.19, 0.27]	0/1000	0.106 / 0.099	0.187 / 0.193
PATT	3	Cor—Cor	0.30	0.26	0.04	[-0.71, 0.76]	0/1000	0.077 / 0.075	0.077 / 0.075
PATT	3	Mis—Cor	0.22	0.18	0.04	[-0.64, 0.68]	0/1000	0.078 / 0.076	0.078 / 0.076
PATT	3	Cor—Mis	2.55	2.78	-0.23	[-1.31, 0.86]	0/1000	0.091 / 0.089	0.096 / 0.095
PATT	3	Mis—Mis	13.73	14.03	-0.30	[-1.40, 0.80]	0/1000	0.093 / 0.090	0.187 / 0.189
PATT	5	Cor—Cor	0.28	0.24	0.04	[-0.68, 0.73]	0/1000	0.075 / 0.074	0.075 / 0.074
PATT	5	Mis—Cor	0.14	0.12	0.02	[-0.56, 0.59]	0/1000	0.076 / 0.074	0.076 / 0.074
PATT	5	Cor—Mis	2.77	2.96	-0.19	[-1.26, 0.87]	0/1000	0.089 / 0.088	0.095 / 0.094
PATT	5	Mis—Mis	13.90	14.08	-0.18	[-1.25, 0.89]	0/1000	0.090 / 0.089	0.187 / 0.189

Relative bias (RB) is 100*(mean estimate - truth)/abs(truth). Absolute relative bias (ARB) is abs(RB), not mean absolute error. ARB reduction is original ARB minus stabilized ARB; positive favors stabilization. MC interval is the protocol's conservative approximate 95% Monte Carlo error interval for bias reduction within each setting. It controls the two mean-bias intervals within that setting; there is no adjustment across the 96 settings. This is not a confidence interval for a stabilized treatment effect. SD denotes Monte Carlo SD. NA means full-run stabilized performance is unavailable because at least one requested estimate failed. No conditional-on-success metric enters these primary outputs. Requested paired replications: 1000.

Table 7: Prospective good overlap: paired point-estimator relative-bias comparison across 1000 sampled data sets.

Estimand	M	PS/PG	Original RB (%)	Stabilized RB (%)	ARB reduction (pp)	MC error interval (pp)	Failed/N	MC SD original/stabilized	RMSE original/stabilized
PATE	1	Cor—Cor	-0.72	NA	NA	NA	6/1000	0.060 / NA	0.060 / NA
PATE	1	Mis—Cor	-0.76	NA	NA	NA	5/1000	0.060 / NA	0.060 / NA
PATE	1	Cor—Mis	-1.19	-0.27	0.92	[-0.25, 1.74]	0/1000	0.068 / 0.075	0.069 / 0.075
PATE	1	Mis—Mis	5.75	6.61	-0.86	[-2.15, 0.43]	0/1000	0.075 / 0.082	0.090 / 0.100
PATE	3	Cor—Cor	-0.58	NA	NA	NA	6/1000	0.057 / NA	0.058 / NA
PATE	3	Mis—Cor	-0.62	NA	NA	NA	5/1000	0.057 / NA	0.057 / NA
PATE	3	Cor—Mis	-0.44	-0.34	0.11	[-0.89, 0.96]	0/1000	0.063 / 0.067	0.063 / 0.067
PATE	3	Mis—Mis	6.37	6.44	-0.06	[-1.26, 1.14]	0/1000	0.072 / 0.075	0.090 / 0.094
PATE	5	Cor—Cor	-0.60	NA	NA	NA	6/1000	0.057 / NA	0.057 / NA
PATE	5	Mis—Cor	-0.62	NA	NA	NA	5/1000	0.057 / NA	0.057 / NA
PATE	5	Cor—Mis	-0.27	-0.23	0.03	[-0.77, 0.78]	0/1000	0.063 / 0.066	0.063 / 0.066
PATE	5	Mis—Mis	6.42	6.41	0.01	[-1.18, 1.20]	0/1000	0.071 / 0.074	0.091 / 0.092
PATT	1	Cor—Cor	-0.51	NA	NA	NA	1/1000	0.067 / NA	0.067 / NA
PATT	1	Mis—Cor	-0.50	-0.52	-0.01	[-0.95, 0.93]	0/1000	0.067 / 0.070	0.067 / 0.070
PATT	1	Cor—Mis	-2.13	-0.74	1.39	[0.37, 2.42]	0/1000	0.073 / 0.077	0.076 / 0.077
PATT	1	Mis—Mis	2.47	3.68	-1.21	[-2.32, -0.09]	0/1000	0.079 / 0.083	0.083 / 0.092
PATT	3	Cor—Cor	-0.59	NA	NA	NA	1/1000	0.062 / NA	0.062 / NA
PATT	3	Mis—Cor	-0.61	-0.60	0.01	[-0.85, 0.86]	0/1000	0.062 / 0.063	0.062 / 0.063
PATT	3	Cor—Mis	-1.31	-0.60	0.71	[-0.23, 1.64]	0/1000	0.067 / 0.070	0.068 / 0.070
PATT	3	Mis—Mis	2.97	3.60	-0.63	[-1.64, 0.38]	0/1000	0.073 / 0.075	0.079 / 0.084
PATT	5	Cor—Cor	-0.64	NA	NA	NA	1/1000	0.061 / NA	0.062 / NA
PATT	5	Mis—Cor	-0.66	-0.65	0.01	[-0.83, 0.85]	0/1000	0.061 / 0.061	0.061 / 0.062
PATT	5	Cor—Mis	-1.02	-0.59	0.43	[-0.48, 1.34]	0/1000	0.065 / 0.067	0.066 / 0.067
PATT	5	Mis—Mis	3.23	3.63	-0.40	[-1.39, 0.59]	0/1000	0.072 / 0.073	0.079 / 0.082

Relative bias (RB) is 100*(mean estimate - truth)/abs(truth). Absolute relative bias (ARB) is abs(RB), not mean absolute error. ARB reduction is original ARB minus stabilized ARB; positive favors stabilization. MC interval is the protocol's conservative approximate 95% Monte Carlo error interval for bias reduction within each setting. It controls the two mean-bias intervals within that setting; there is no adjustment across the 96 settings. This is not a confidence interval for a stabilized treatment effect. SD denotes Monte Carlo SD. NA means full-run stabilized performance is unavailable because at least one requested estimate failed. No conditional-on-success metric enters these primary outputs. Requested paired replications: 1000.

Table 8: Prospective poor overlap: paired point-estimator relative-bias comparison across 1000 sampled data sets.

Estimand	M	PS/PG	Original RB (%)	Stabilized RB (%)	ARB reduction (pp)	MC error interval (pp)	Failed/N	MC SD original/stabilized	RMSE original/stabilized
PATE	1	Cor—Cor	-1.03	NA	NA	NA	7/1000	0.116 / NA	0.117 / NA
PATE	1	Mis—Cor	-1.19	NA	NA	NA	5/1000	0.118 / NA	0.118 / NA
PATE	1	Cor—Mis	-0.64	0.80	-0.16	[-2.10, 1.78]	0/1000	0.139 / 0.158	0.139 / 0.158
PATE	1	Mis—Mis	23.87	25.15	-1.28	[-3.93, 1.37]	0/1000	0.152 / 0.172	0.257 / 0.277
PATE	3	Cor—Cor	-0.32	NA	NA	NA	7/1000	0.107 / NA	0.107 / NA
PATE	3	Mis—Cor	-0.43	NA	NA	NA	5/1000	0.108 / NA	0.108 / NA
PATE	3	Cor—Mis	1.79	1.99	-0.19	[-2.35, 1.96]	0/1000	0.126 / 0.137	0.127 / 0.138
PATE	3	Mis—Mis	25.53	25.56	-0.03	[-2.44, 2.38]	0/1000	0.142 / 0.152	0.263 / 0.269
PATE	5	Cor—Cor	-0.02	NA	NA	NA	7/1000	0.104 / NA	0.104 / NA
PATE	5	Mis—Cor	-0.14	NA	NA	NA	5/1000	0.104 / NA	0.104 / NA
PATE	5	Cor—Mis	2.71	2.69	0.01	[-2.05, 2.08]	0/1000	0.123 / 0.129	0.125 / 0.131
PATE	5	Mis—Mis	25.80	25.69	0.11	[-2.23, 2.45]	0/1000	0.140 / 0.145	0.264 / 0.266
PATT	1	Cor—Cor	-0.67	-0.55	0.12	[-1.19, 1.29]	0/1000	0.103 / 0.115	0.103 / 0.115
PATT	1	Mis—Cor	-0.62	-0.53	0.09	[-1.20, 1.24]	0/1000	0.103 / 0.113	0.103 / 0.113
PATT	1	Cor—Mis	-0.84	1.24	-0.39	[-2.02, 1.24]	0/1000	0.129 / 0.143	0.129 / 0.143
PATT	1	Mis—Mis	11.20	12.75	-1.55	[-3.16, 0.07]	0/1000	0.128 / 0.141	0.184 / 0.206
PATT	3	Cor—Cor	-0.35	-0.29	0.06	[-0.91, 0.93]	0/1000	0.096 / 0.103	0.096 / 0.103
PATT	3	Mis—Cor	-0.47	-0.39	0.08	[-1.00, 1.04]	0/1000	0.096 / 0.103	0.097 / 0.103
PATT	3	Cor—Mis	0.95	1.86	-0.91	[-2.38, 0.56]	0/1000	0.117 / 0.127	0.118 / 0.129
PATT	3	Mis—Mis	12.27	12.96	-0.69	[-2.16, 0.78]	0/1000	0.118 / 0.127	0.187 / 0.199
PATT	5	Cor—Cor	-0.26	-0.22	0.04	[-0.81, 0.83]	0/1000	0.095 / 0.099	0.095 / 0.099
PATT	5	Mis—Cor	-0.38	-0.32	0.06	[-0.92, 0.95]	0/1000	0.096 / 0.100	0.096 / 0.100
PATT	5	Cor—Mis	1.58	2.11	-0.53	[-1.95, 0.89]	0/1000	0.115 / 0.122	0.117 / 0.124
PATT	5	Mis—Mis	12.58	13.00	-0.42	[-1.86, 1.02]	0/1000	0.117 / 0.122	0.189 / 0.196

Relative bias (RB) is 100*(mean estimate - truth)/abs(truth). Absolute relative bias (ARB) is abs(RB), not mean absolute error. ARB reduction is original ARB minus stabilized ARB; positive favors stabilization. MC interval is the protocol's conservative approximate 95% Monte Carlo error interval for bias reduction within each setting. It controls the two mean-bias intervals within that setting; there is no adjustment across the 96 settings. This is not a confidence interval for a stabilized treatment effect. SD denotes Monte Carlo SD. NA means full-run stabilized performance is unavailable because at least one requested estimate failed. No conditional-on-success metric enters these primary outputs. Requested paired replications: 1000.